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TECHNICAL SUMMARY

Ultra-High-Speed PMSM Drive Development

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ABSTRACT

Delay-aware multivariable control for a GaN-based PMSM drive designed for 100,000 rpm, run to 95,000 rpm on hardware. This project builds a complete ultra-high-speed PMSM drive and develops multivariable (MIMO) control that accounts for delay, so the drive stays stable and responsive at very high speed. The drive uses a GaN-based inverter feeding a Maxon motor with Hall-sensor feedback, controlled by FOC-based strategies on a TI C2000 (F28379D) LaunchPad. Control is designed in MATLAB, validated in real time on a PLECS RT Box FPGA model, and then deployed as embedded code.

Design speed	100,000 rpm
Reached on hardware	95,000 rpm
Speed margin (MIMO control)	+188%
Inverter technology	GaN

1. Problem and motivation

At 100,000 rpm the motor's electrical frequency is in the kilohertz range, and the controller gets only a few samples per electrical cycle. Computation and PWM delays then become a significant fraction of the cycle and couple the d- and q-axis currents. Conventional PI current control loses stability margin exactly where the drive needs it most.

Ultra-high-speed motors make compressors and turbomachinery smaller and lighter, but only if the drive stays stable and responsive at full speed.

2. System description

- GaN-based inverter
- Maxon PMSM with Hall-sensor feedback
- TI C2000 F28379D LaunchPad controller
- PLECS RT Box for real-time HIL

3. Methodology

- Designed FOC-based control strategies for operation up to 100,000 rpm

- Developed and analyzed MIMO control techniques that mitigate delay effects and improve high-speed stability and dynamic performance
- Validated state-of-the-art and proposed methods in real time on a PLECS RT Box FPGA model
- Implemented the algorithms as embedded code on the microcontroller and moved from simulation to physical hardware

4. Figures

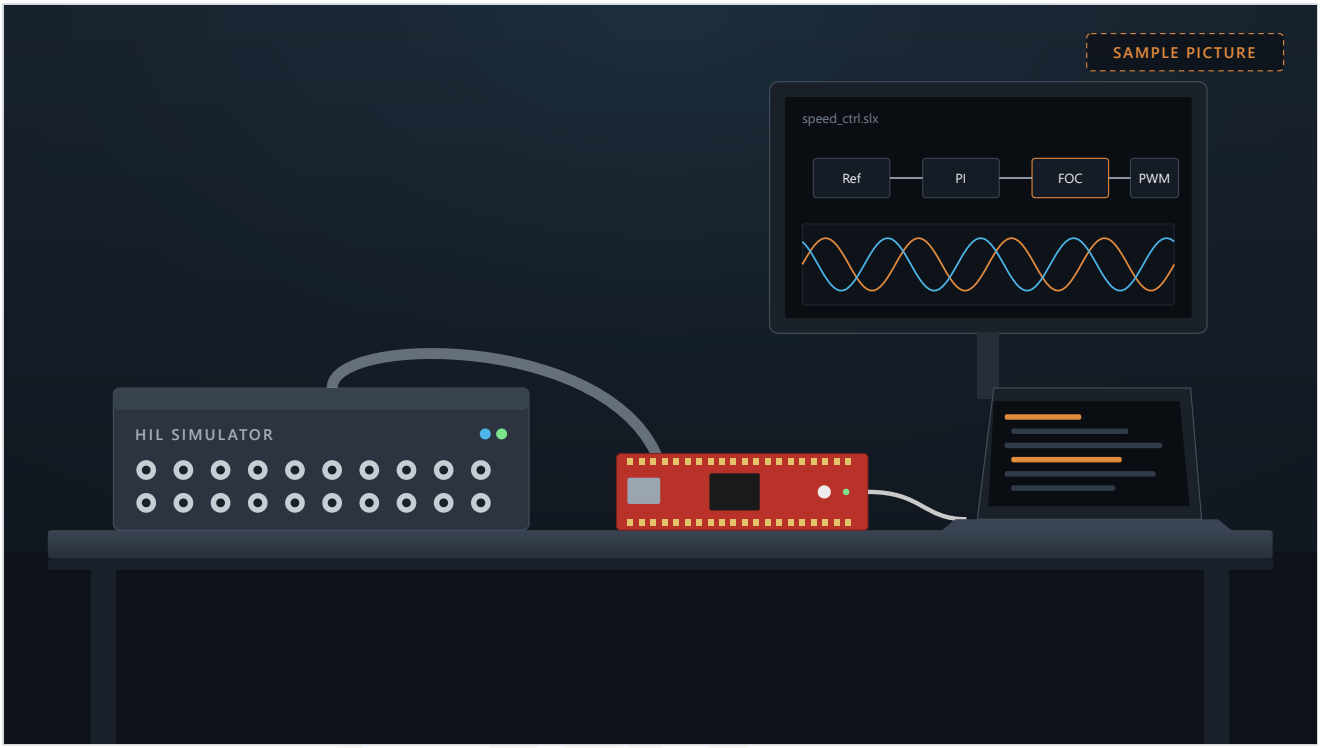


Figure 1. HIL setup: PLECS RT Box, TI C2000 LaunchPad running the control firmware, and the model on screen.

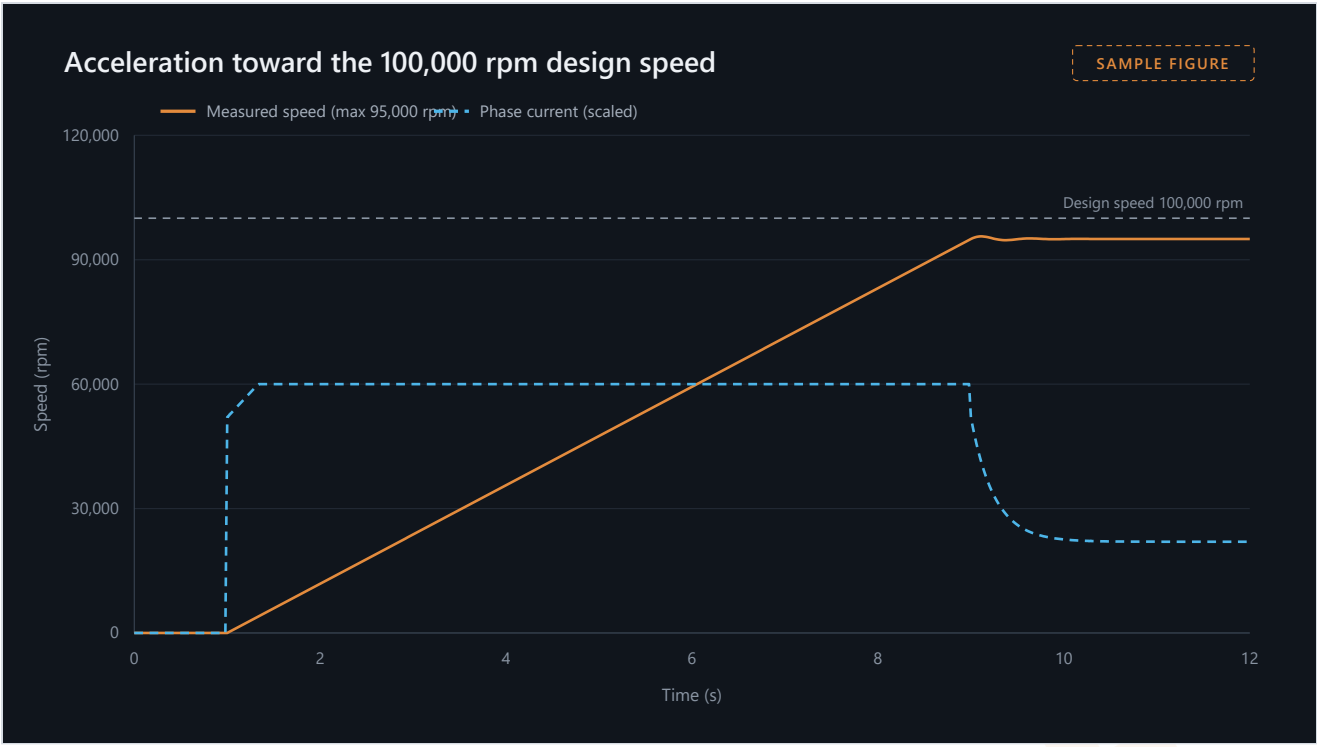


Figure 2. Acceleration toward the 100,000 rpm design speed; 95,000 rpm reached on hardware.

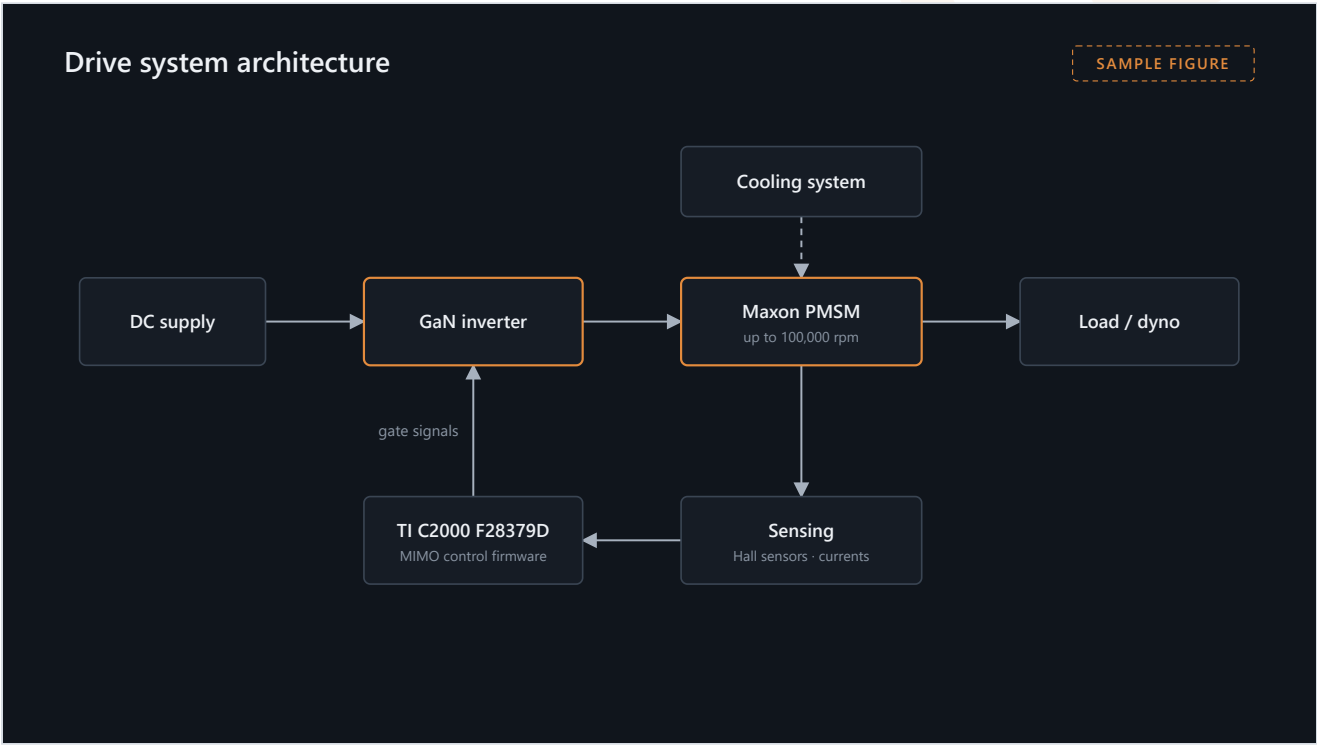


Figure 3. Drive system: GaN inverter, Maxon PMSM with Hall sensors, TI C2000 F28379D controller.

5. Results

- Delay-aware MIMO control improved the speed margin by 188%
- Reached a maximum speed of 95,000 rpm on hardware
- Robust multivariable current control approach submitted to IEEE Transactions on Industrial Electronics (under review)

6. Team and contribution

Research with M. Karimi-Ghartemani and S. Choi at Mississippi State University, co-authors on the related publications.

Contribution: Designs the control strategies, including delay-aware MIMO current control, builds the drive system, runs HIL validation on PLECS RT Box, and implements the algorithms as embedded code in Code Composer Studio.

Tools: MATLAB/Simulink, PLECS, RT Box, TI C2000 F28379D, Code Composer Studio, GaN inverter

7. References

1. SSCET 2026 paper — PI speed controllers (DOI): https://doi.org/10.1109/SSCET69323.2026.11735668

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